

Can AI make a robot dance to music?

Human dance generation is expressive; humanoid execution is constrained by physics.

PINNED SHOWCASE: expressive visual demo



FREE-ROOT STABLE: physics rollout



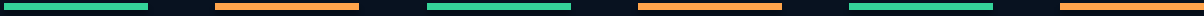
SHOWCASE

looks like dancing

FREE-ROOT

tests physical survival

Key question: not only "does it move?", but "can it stand?"



The missing link

BACKGROUND | MUSIC-TO-DANCE VS ROBOT EXECUTION

Music-to-Dance

EDGE / AIST++ / Bailando

Generates human skeleton or SMPL motion.
Great for visual choreography,
not robot dynamics.

Our focus

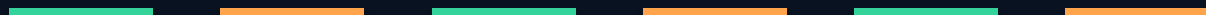
Physics-aware retargeting from
music-conditioned human
dance
to Unitree G1 rollout.

Robot Dance

MoCap / choreography / MPC / RL

Can execute designed motions,
but usually not direct arbitrary
music-conditioned generation.

Direct music → stable humanoid dance is not yet an off-the-shelf solution.



Project goal and innovation

POSITIONING | A PHYSICS-AWARE BRIDGE

Input

Music / existing
generated pkl
SMPL human dance
motion

Our contribution

- **Retarget**
SMPL-to-G1 mapping and diagnostics
- **Constrain**
safe / expressive / balance / showcase profiles
- **Evaluate**
pinned visual check + free-root physics rollout

Output

PINNED

visual showcase

FREE-ROOT

stable rollout

Innovation: an honest bridge and evaluation framework, not just a video demo.

Pipeline: retarget, constrain, simulate

METHOD | FROM SMPL DANCE TO G1 ROLLOUT

01

**Music / pkl
motion**

Generated or existing
dance clips.

02

**SMPL human
dance**

Human motion prior
from music models.

03

**SMPL-to-G1
retargeting**

Map body parts and
adapt arms.

04

Motion profiles

Safe, expressive,
balance, showcase.

05

**MuJoCo PD
rollout**

Pinned display or free-
root validation.

Contact-aware balance correction + support-foot locking for the free-root branch

Experimental setup

EVALUATION | WHAT WE TESTED

ROBOT

Unitree G1 in MuJoCo

CONTROLLER

PD tracking

MOTION CLIPS

Pop-style generated motion + AIST++ Waack

MODES

Pinned-root showcase vs free-root rollout

METRICS

Stability, fall time, pelvis height, amplitude, foot slip

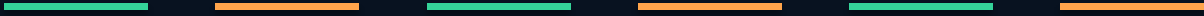
Pinned showcase

Fixed root
Best visual expressiveness
Not a physical proof

Free-root stable

Floating root
Normal gravity simulation
Physical stability check

This separation avoids overclaiming: visual quality and physical validity are different questions.



Results: expressive motions fall, stable

ANALYSIS | STABILITY-EXPRESSIVENESS TRADE-OFF

motions shrink

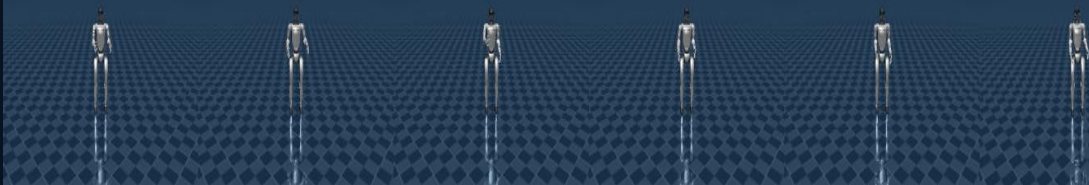
PINNED SHOWCASE: expressive visual demo

0.5s 2.0s 3.5s 5.0s 6.5s 8.0s



FREE-ROOT STABLE: physics rollout

0.5s 2.0s 3.5s 5.0s 6.5s 8.0s



Expressive free-root

can fall early

Contact-aware / footlock

100% stable on tested clips

Pinned showcase

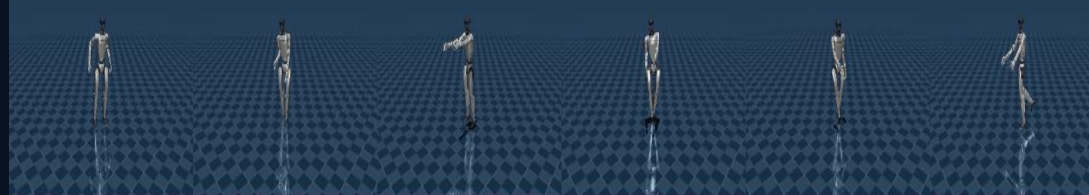
best visual dance quality

Foot slip

small reduction after lock

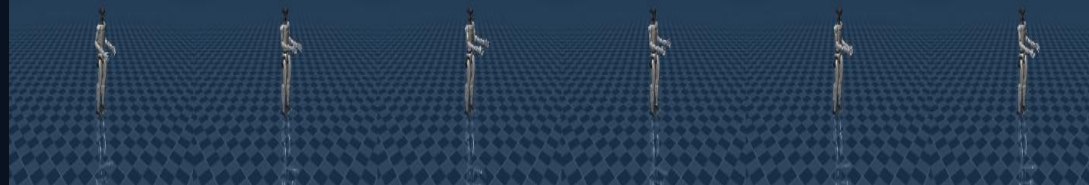
PINNED SHOWCASE: expressive visual demo

0.5s 2.5s 4.5s 6.5s 8.5s 10.5s



FREE-ROOT STABLE: physics rollout

0.5s 2.5s 4.5s 6.5s 8.5s 10.5s



Pinned answers: does it look like dancing? Free-root answers: can it physically survive?

Takeaways and next steps

CONCLUSION | WHAT WE LEARNED

Key takeaways

- **Human dance is not robot motion**
Music-to-dance output still needs robot-aware processing.
- **Trade-off is central**
Expressiveness and free-root stability pull in opposite directions.
- **Two outputs are honest**
Showcase for visual quality; free-root rollout for physics.

Future work

- Contact-aware lower-body IK
- COM-aware root correction
- Better support-phase detection
- Learning-based policy tracking
- More music styles and longer clips

Next goal: not only keep G1 standing, but preserve dance expressiveness under real physical constraints.

